

Project Design Phase
Proposed Solution Template

Date	18 June 2026
Team ID	
Project Name	ShopEZ-E-commerce-Application
Maximum Marks	2 Marks

Proposed Solution Template:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Consumers need a reliable, user-friendly platform to discover, compare, and purchase a wide variety of products online from the comfort of their homes. Simultaneously, small and medium sellers struggle to reach a broader digital audience without an accessible, robust digital storefront.
2.	Idea / Solution description	A full-stack MERN (MongoDB, Express.js, React.js, Node.js) e-commerce marketplace that connects buyers and sellers. The application features a comprehensive product catalog with search/filter options, secure user authentication, shopping cart functionality, a seamless checkout process, and an admin/seller dashboard for inventory and order management.
3.	Novelty / Uniqueness	While e-commerce is an established industry, ShopEZ focuses on a highly optimized, lightweight MERN architecture that guarantees lightning-fast page loads and a highly intuitive, mobile-first user interface, ensuring a frictionless shopping experience even for users on lower-end devices or slower networks.
4.	Social Impact / Customer Satisfaction	The platform maximizes customer convenience by bringing a diverse range of products directly to their fingertips. Furthermore, it empowers local retailers

		and small business owners by providing them with an accessible digital marketplace to scale their sales and reach new demographics.
5.	Business Model (Revenue Model)	Commission & Advertising: The primary revenue stream is a commission-based model, where the platform takes a small percentage fee on every successful transaction. Secondary revenue can be generated by allowing sellers to pay for featured product placements or banner advertisements on the homepage.
6.	Scalability of the Solution	The MERN stack's modular nature allows for highly efficient scaling. MongoDB's NoSQL structure easily accommodates a massive, ever-growing product catalog. The Express/Node backend can be scaled horizontally to manage high traffic during major sale events, while the React frontend ensures smooth, dynamic rendering regardless of the database size.